# Institute of Technology of Cambodia

## Department of Civil Engineering

# Reinforcement Concrete

# Project 01

**Lecture : Dr. Sambath KY**

| | | |
|---|---|---|
| **Name** | **:** | **Team** |
| **ID** | **:** | **e2022—-** |
| **Group** | **:** | **I4-GCI-A1** |

## 2025-2026

*Note that*: it just sample, maybe Makara is the one who design the cover.

# Reinforcement Concrete

# Project 01

We can delete this since it no need

# Table of Contents


1 Introduction . . . . . 1

2 Team Organization . . . . . 2

3 Meeting Minutes & Progress Reports . . . . . 3

4 Architectural & Structural Assumptions . . . . . 4

5 Design & Optimization Process . . . . . 5

6 Detailed Design Calculations . . . . . 6

6.1 Load . . . . . 6

7 Design Drawings . . . . . 8

8 Results, Discussion & Conclusion . . . . . 9

References . . . . . 10

Appendices . . . . . 11

# List of Figures

# List of Tables

Table 6.1: Impose loads on floors by EN 1991-1-1:2002 . . . . . 7

# 1 Introduction

This is Our Team Intro.

# 2 Team Organization

# 3 Meeting Minutes & Progress Reports

# 4 Architectural & Structural Assumptions

This is Our Team Assumption.

# 5 Design & Optimization Process

This is Design

# 6 Detailed Design Calculations

## 6.1 Load

For imposed loads on floors according to Table 6.2 and 6.4 in EN 1991-1-1:2002 [3]:

| Category | Sub-category / Use | $q_k$ [kN/m²] | $Q_k$ [kN] |
|---|---|---|---|
| **Category A** | Areas for domestic and residential activities | | |
| | - Floors | 1,5 to 2,0 | 2,0 to 3,0 |
| | - Stairs | 2,0 to 4,0 | 2,0 to 4,0 |
| | - Balconies | 2,5 to 4,0 | 2,0 to 3,0 |
| **Category B** | Office areas | 2,0 to 3,0 | 1,5 to 4,5 |
| **Category C** | Areas where people may congregate | | |
| C1 | Areas with tables (e.g. schools, cafes, restaurants, reading rooms) | 2,0 to 3,0 | 3,0 to 4,0 |
| C2 | Areas with fixed seats (e.g. theatres, cinemas, churches) | 3,0 to 4,0 | 2,5 to 7,0 (4,0) |
| C3 | Areas without obstacles for moving people (e.g. museums, entrance halls) | 3,0 to 5,0 | 4,0 to 7,0 |
| C4 | Areas with possible physical activities (e.g. dance halls, gyms) | 3,5 to 5,0 | 3,5 to 7,0 |
| C5 | Areas susceptible to large crowds (e.g. concert halls, sports halls with stands) | 5,0 to 7,5 | 3,5 to 4,5 |
| **Category D** | Shopping areas | | |
| D1 | Areas in general retail shops | 4,0 to 5,0 | 3,5 to 7,0 (4,0) |
| D2 | Areas in department stores | 4,0 to 5,0 | 3,5 to 7,0 |

| | | | |
|---|---|---|---|
| **Category E** | Storage and industrial activities (general storage, accessible areas) | 7,5 | 7,0 |

Table 6.1: Impose loads on floors by EN 1991-1-1:2002

# 7 Design Drawings

This is Our Team Design Drawings.

# 8 Results, Discussion & Conclusion

This is Our Team Conclusion.

# References

[1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.

[2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.

[3] British Standards Institution, *BS EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, Incorporating corrigenda December 2004 and March 2009, BSI, 2002.

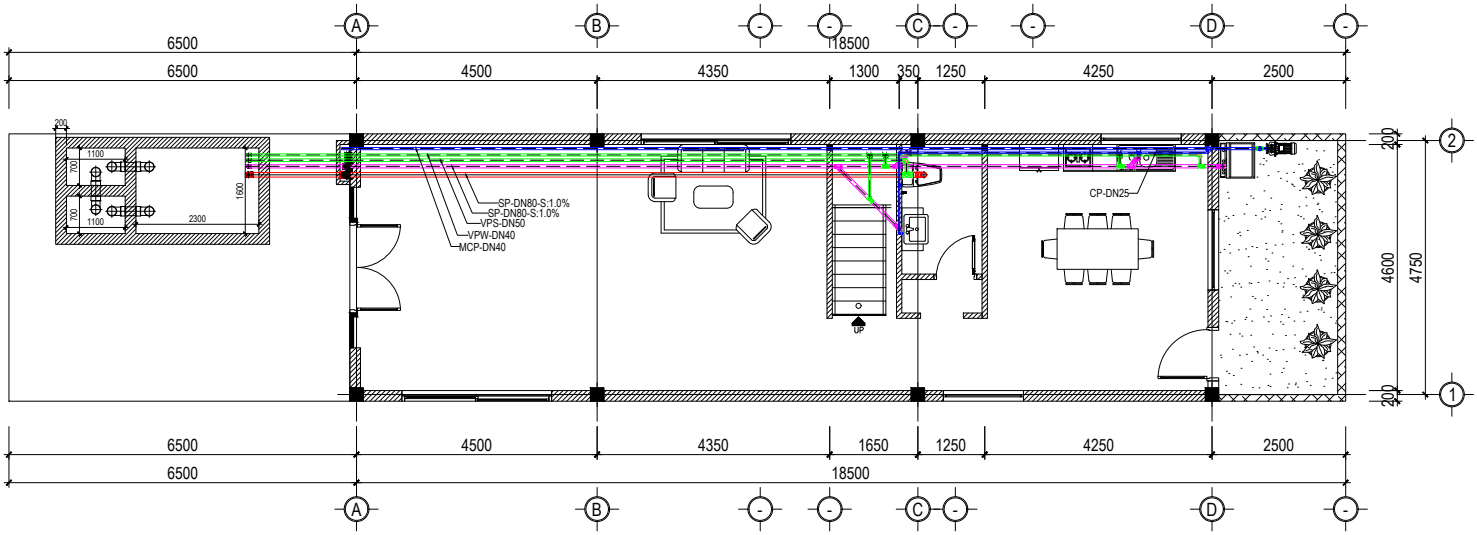
[4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.

[5] CEN, *Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, EN 1991-1-4:2005+A1, English version, April 2010.

*Note that :* We may use the referennces provide by professor (check in telegram group) if it still not enought we can add more maybe.

# Appendices

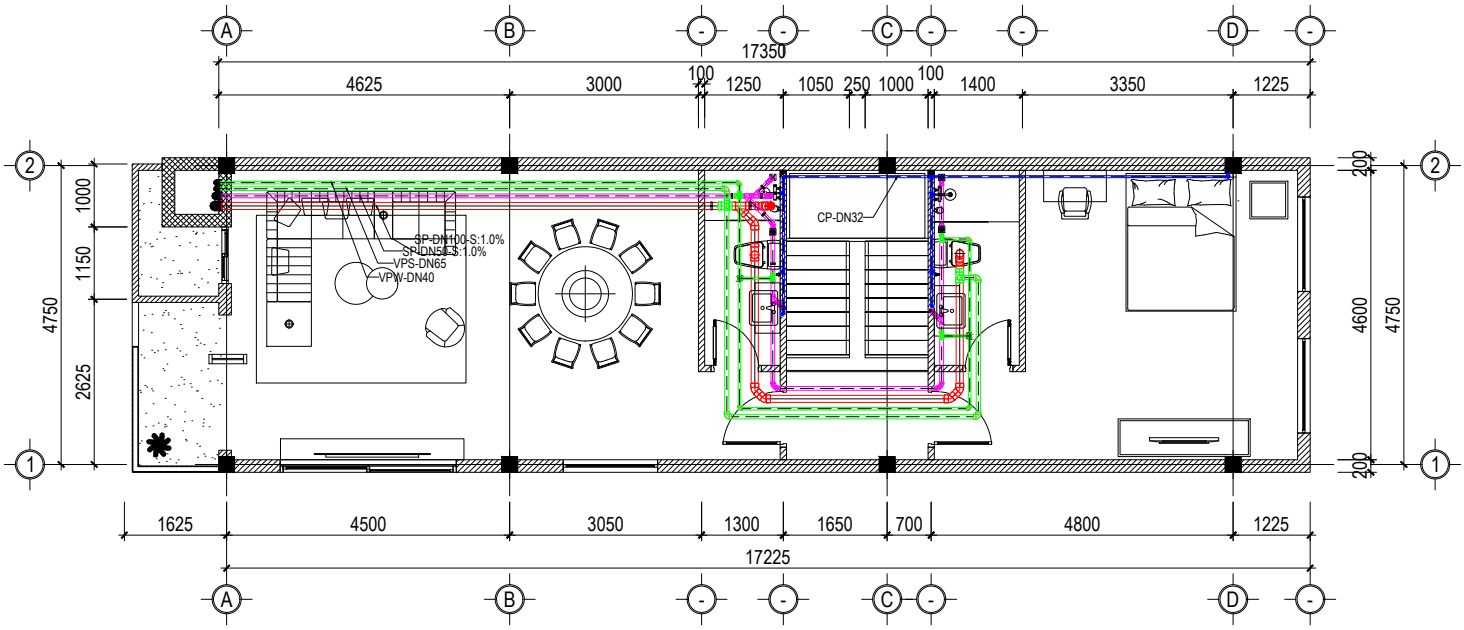
This is appendices




GROUND FLOOR
SCALE : 1/100

| GROUND FLOOR | | |
|---|---|---|
| SP | 3" | 80mm |
| WP | 2" | 50mm |
| VSP | 2" | 50mm |
| VWP | 1.5" | 40mm |
| CP | 1.5" | 25mm |

| | | |
|---|---|---|
| Course: Installation and Sanatary<br>Lecturer: MAY Raksmey | Student: CHUCH Manet<br>ID: e20220847 | Plumbing Plan |




FIRST FLOOR
SCALE : 1/85

| 1ST FLOOR | | |
|---|---|---|
| SP | 4" | 100mm |
| WP | 2" | 50mm |
| VSP | 2.5" | 65mm |
| VWP | 1.5" | 40mm |
| CP | 1.5" | 32mm |

| | | |
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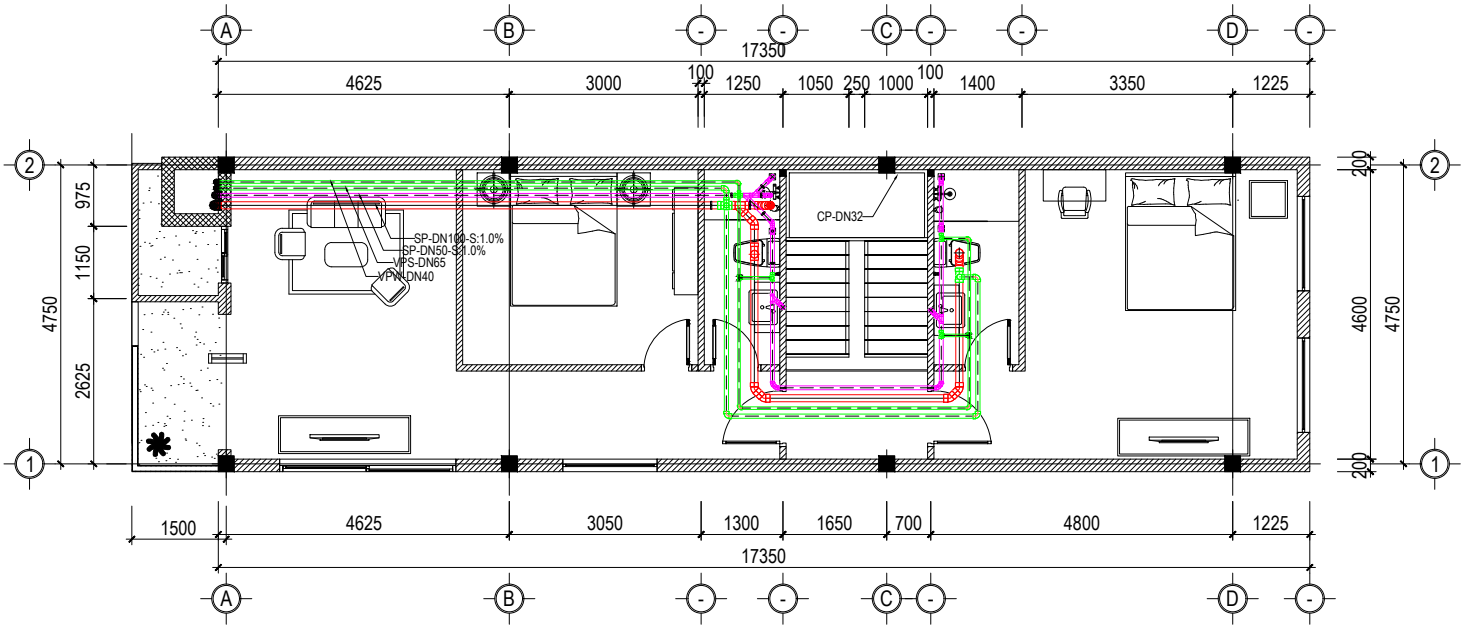


SECOND FLOOR
SCALE : 1/85

| 2ND FLOOR | | |
|---|---|---|
| SP | 4" | 100mm |
| WP | 2" | 50mm |
| VSP | 2.5" | 65mm |
| VWP | 1.5" | 40mm |
| CP | 1.5" | 32mm |

| | | |
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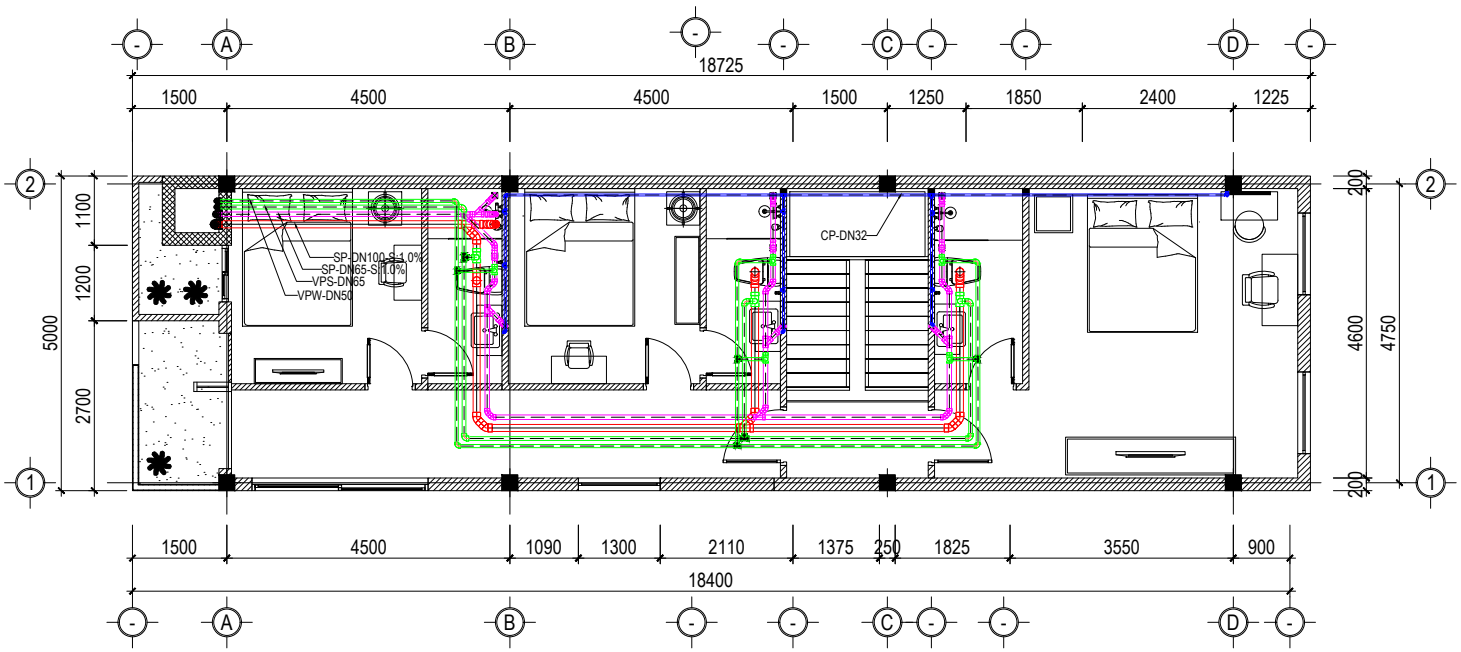


THIRD FLOOR
SCALE : 1/85

| 3RD FLOOR | | |
|---|---|---|
| SP | 4" | 100mm |
| WP | 2.5" | 65mm |
| VSP | 2.5" | 65mm |
| VWP | 2" | 50mm |
| CP | 1.5" | 32mm |

| | | |
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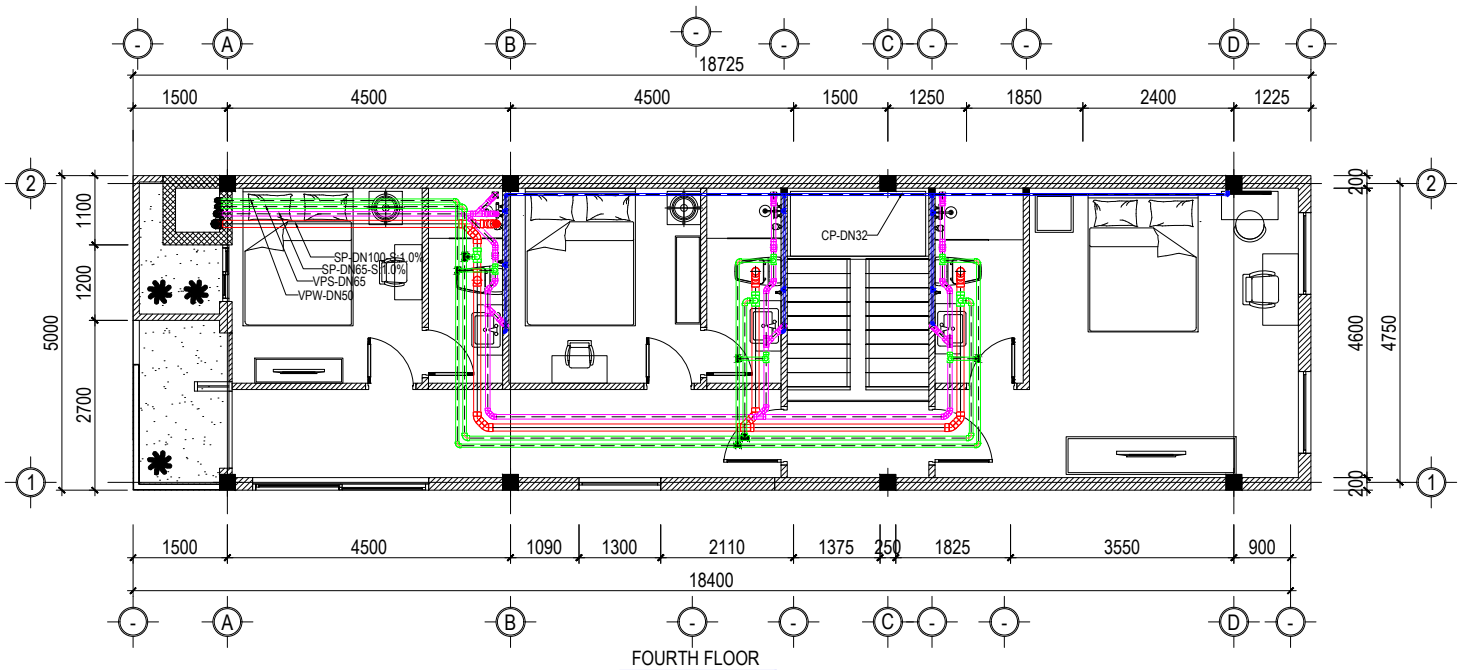


FOURTH FLOOR

SCALE : 1/85

| 3RD FLOOR | | |
|---|---|---|
| SP | 4" | 100mm |
| WP | 2.5" | 65mm |
| VSP | 2.5" | 65mm |
| VWP | 2" | 50mm |
| CP | 1.5" | 32mm |

| | | |
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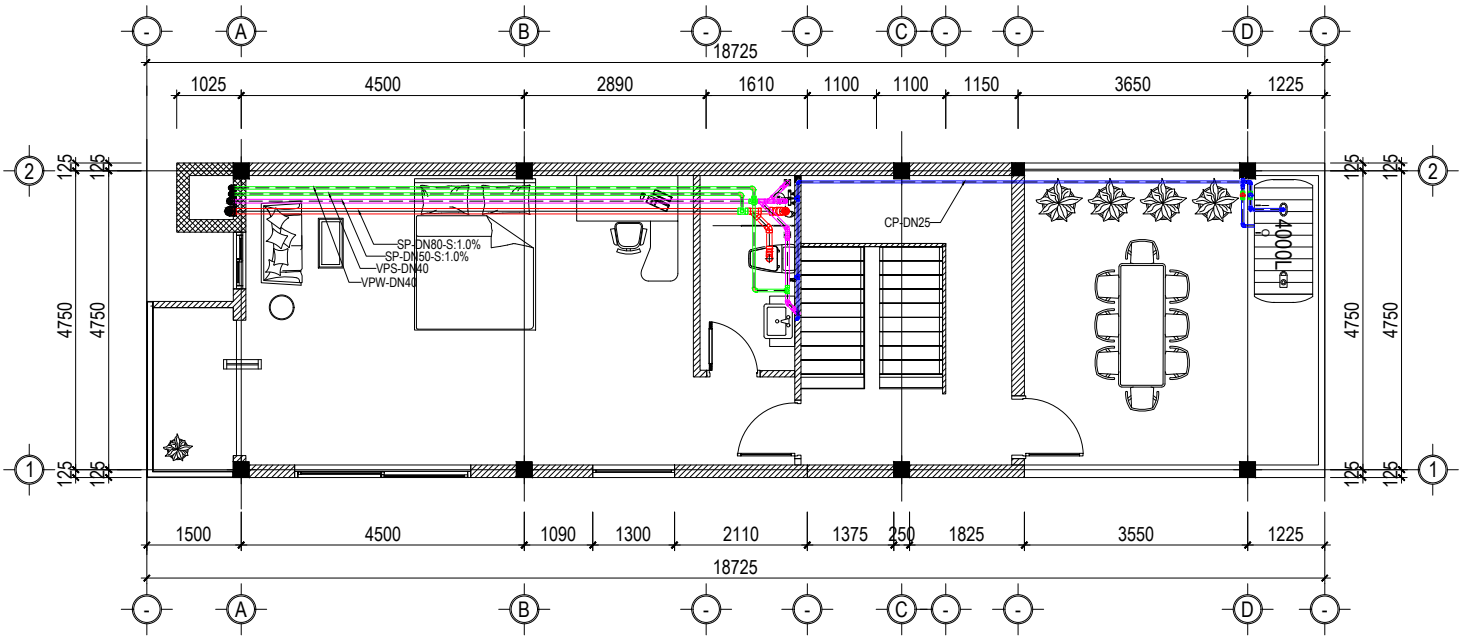


TERRACE FLOOR

SCALE : 1/85

| GROUND FLOOR | | |
|---|---|---|
| SP | 3" | 80mm |
| WP | 2" | 50mm |
| VSP | 1.5" | 40mm |
| VWP | 1.5" | 40mm |
| CP | 1.5" | 25mm |

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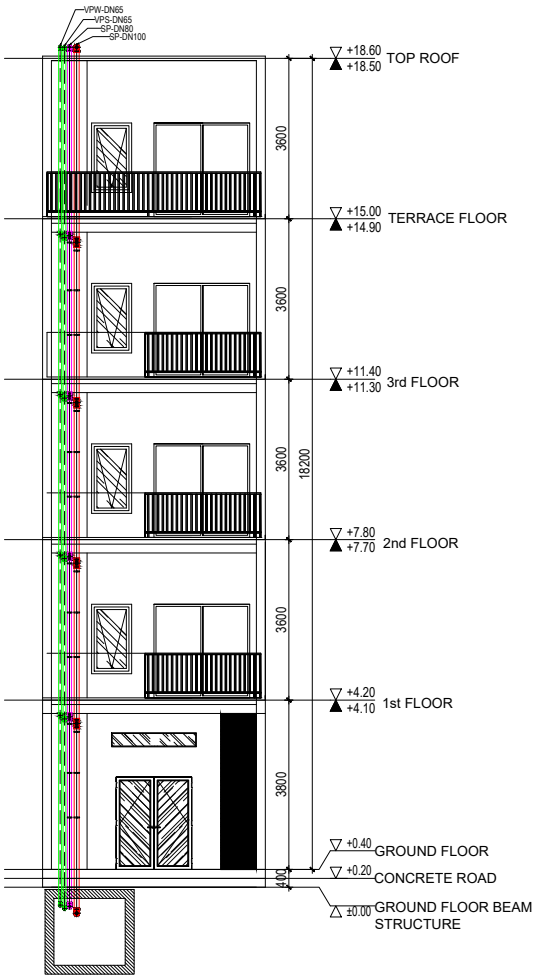


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This is update of appendices